

Saptarshi Neil Sinha, Ph.D.

Computer Graphics and AI Scientist

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Professional Summary

Profile I am a 3D vision and perception scientist with a Ph.D. from TU Darmstadt and 13 years of experience in research and industry, from real-time rendering of large 3D models in the browser to learning-based 3D scene digitization. My current work covers perception beyond RGB cameras, where I train models on synthetic data for object detection on the edge, surface defect detection and pose estimation, as well as multi-modal CAD editing with large language models and agentic systems. As senior research scientist and technology lead at Fraunhofer IGD, I have delivered EU and national research projects, shipped rendering projects with BMW and Mercedes, supervised 15+ theses and taught at TU Darmstadt.

Skills

Programming	C++ (expert), TypeScript (expert), JavaScript (expert), Python (advanced), Java (advanced), SQL (advanced), CUDA (proficient)
AI & Vision Methods	3D reconstruction, 3D Gaussian splatting, object detection, semantic segmentation, material estimation, inpainting and neural image restoration, synthetic data pipelines, cross-modal learning, shape retrieval, edge AI, MLOps, large language models (fine-tuning, RAG, agentic AI)
Research Interests	Multi-modal scene understanding and realistic scene synthesis, scene digitization with spectral and event cameras, learning-based scene editing and analysis, real-time rendering of large models in the web, multi-modal CAD editing
Domains	Automotive and OEM engineering, smart farming and robotics, cultural heritage, content and creative industries
Leadership	Technical and scientific lead, work package leader, project scientific board member, grant writing, thesis supervision, university teaching, peer review
Languages	Bengali (native), English (fluent), German (fluent), Hindi (basic)

Employment History

Sep 2026 – Present **Guest Researcher**, TU Delft, Computer Graphics and Visualization, Delft, Netherlands
Works on neural scene representations, multi-spectral 3D scene analysis and learning-based spectral CT digitization via Gaussian splatting with basis material decomposition.
Collaborates on learning-based approach allowing efficient multi-modal analysis of multi-spectral painting dataset
Co-supervises master and bachelor students with Prof. Michael Weinmann.

Employment History (continued)

- Nov 2025 – Sep 2026  **Senior Research Scientist and Technology Lead**, Fraunhofer IGD, Virtual and Augmented Reality, Darmstadt
Set the technical direction for 3D vision, secured two Horizon Europe grants (incl. COLOURS), led technology adaptation work packages and represented the institute before the European Commission.
Built agentic AI systems coupling LLMs with multi-modal CAD data, published on edge-based perception and procedural synthetic data (ScratchSim, ECCV Workshop 2026), and mentored junior researchers, a Fulbright scholar and thesis students.
- Mar 2023 – Oct 2025  **Research Associate**, Fraunhofer IGD, Virtual and Augmented Reality, Darmstadt
Led the AI core and image-based rendering work packages in the EU project PERCEIVE and the SINATRA smart farming project on edge AI and purely synthetic 6D pose estimation.
Pursued a Ph.D. in parallel on spectral and semantic Gaussian splatting (ISPRS J. Photogramm. Remote Sens.), built a synthetic-data restoration pipeline for historical autochromes exhibited at the University Museum Trondheim, and co-deployed the institute GPU cluster.
- Feb 2019 – Feb 2023  **Research Engineer**, Fraunhofer IGD, Visual Computing System Technologies, Darmstadt
Delivered real-time browser-based rendering of large CAD models for automotive customers, including screen-space level-of-detail selection with foveated rendering, depth-image-based and iterative rendering, and WebVR in Instant3DHub (spin-off Threedy).
Shipped production rendering features for Daimler, built collaborative client demos, and cut initial load time for large assemblies by at least 10% through out-of-core streaming and multi-draw.
- Mar 2016 – Feb 2019  **Senior Software Developer**, IMC AG, Freiburg
Developed new modules for the Content Studio and Process Guide e-learning products.
Built the Process Guide backend in Java with Maven and Hibernate and the Content Studio frontend components with Qt.
- May 2013 – Feb 2016  **Junior Software Developer**, IMC AG, Saarbrücken
Implemented frontend and backend features for e-learning products alongside the Master studies.
Automated the functional test environment using Selenium.

Education

- Dec 2022 – Nov 2025  **Ph.D. in Computer Science**, TU Darmstadt
Thesis: *Scene Understanding Meets Realistic Scene Synthesis: Novel Learning-based Technologies for Scene Digitization, Analysis and Editing*
Supervisors: Prof. Dr. Arjan Kuijper, Prof. Dr. Michael Weinmann
- Apr 2013 – Feb 2016  **M.Sc. in Computer Science**, Saarland University
Thesis: *Parameter Analysis of Screen Space Ambient Occlusion Methods*
Supervisor: Prof. Dr. Philipp Slusallek
- Aug 2007 – Jul 2011  **B.Tech. in Computer Science & Engineering**, West Bengal University of Technology (WBUT)
Thesis: *Cryptanalysis of Knapsack Cipher using Evolutionary Algorithms*
Supervisor: Prof. Malay Kule

Projects, Teaching and Outreach

Projects

- September 2025 – Present  Collaborates on research project related neural scene understanding, grant and proposal writing.
- October 2025 – September 2026  Leads the work package *Technology Adaptation and Development* in the Horizon Europe project **COLOURS**.
- March 2023 – February 2026  Leads the work package *AI Core and Image-based Rendering* in the Horizon Europe project **PERCEIVE** and coordinates partners across Europe.
- October 2023 – September 2024  Managed **SINATRA**, a smart farming research project with the German Federal Ministry of Agriculture, including edge AI for robotic platforms.
- April 2022 – February 2023  Co-designed and helped deploy the Fraunhofer IGD GPU cluster for faster AI training and inference.
- March 2019 – February 2023  Optimized rendering in *Instant3DHub*, now commercialized as the spin-off Threedy.
- March 2016 – February 2019  Developed features for the Content Studio and Process Guide products at IMC AG.

Supervision and Teaching

- 2022 – Present  Teaches *Material Asset Representation, Segmentation and Generation from Images using Learning-based Approaches* at TU Darmstadt, including course material and exams.
-  Supervises Master and Bachelor thesis students at TU Darmstadt and TU Delft
- 2023 – Present  Leads researcher teams as work package lead and scientific board member in Horizon Europe projects, and presents at mid-term and final project reviews.
- 2024 – Present  Reviews for the ISPRS Journal of Photogrammetry and Remote Sensing, ISPRS annals, Pattern Recognition, ECCV, and the ACM Journal on Computing and Cultural Heritage.

Invited Talks, Workshops and Exhibitions

- 2025  Invited talk on *Neural Restoration of Greening Defects in Digitized Autochromes using Purely Synthetic Data*, Fragile Colours, Victoria and Albert Museum, London, UK.
-  Work on digital restoration of autochromes shown at the exhibition *Fragile Colours*, University Museum Trondheim, Norway.
- 2024  Organized the workshop *Enhancing Perceptive Reconstructions in 3D using Deep Learning*, Web3D 2024, Guimarães, Portugal.
-  Invited talk on *Artificial Intelligence and Reconstruction of Coloured Art Objects*, InArt 2024, Oslo, Norway.

Selected Publications

Learning-based scene digitization

JOURNAL · 2025

SpectralGaussians: Semantic, spectral 3D Gaussian splatting for multi-spectral scene representation, visualization and analysis

S. N. Sinha, H. Graf, M. Weinmann

ISPRS Journal of Photogrammetry and Remote Sensing, vol. 227, pp. 789–803

[Paper](#)

ECCV WORKSHOPS · 2024

MACGaussian: Robust 3D Gaussian splatting from sparse input views using high-precision measurement-arm-camera (MAC) capture

S. N. Sinha, M. A. Shahid, M. Weinmann

Computer Vision — ECCV 2024 Workshops, Springer Nature Switzerland

[Paper](#)

WEB3D · 2024

SpectralSplatsViewer: An interactive web-based tool for visualizing cross-spectral Gaussian splats

S. N. Sinha, J. Kühn, H. Graf, M. Weinmann

Proceedings of the 29th International ACM Conference on 3D Web Technology (Web3D), ACM

[Paper](#)

ECCV WORKSHOP (TWINWORLD) · 2026

OmniDreamer: Disocclusion-aware dynamic view sampling for panoramic 3D scene completion

A. Agrawal, **S. N. Sinha**, F. Rücker

Workshop paper

[Paper](#)

Curated synthetic data for task-specific scene understanding

ECCV WORKSHOPS · 2026

ScratchSim: A procedural synthetic data pipeline for surface scratch detection

P. J. Kühn*, **S. N. Sinha***, T. Kleist, R. Hoffmann, A. Kuijper, M. Weinmann

*Accepted at ECCV Workshop 2026 (CDEL) · *equal contribution*

[Paper](#) [Project page](#)

GCPR · 2026

SynSur: An end-to-end generative pipeline for synthetic industrial surface defect generation and detection

P. J. Kühn, M. Pommeranz, A. Kuijper, **S. N. Sinha**

Accepted at GCPR 2026

[Paper](#)

JOURNAL · 2026

6D strawberry pose estimation: Real-time and edge AI solutions using purely synthetic training data

S. N. Sinha, P. J. Kühn, M. S. Goschke, M. Weinmann

ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences, vol. XI-2-2026, pp. 751–758

[Paper](#)

PRE-PRINT · UNDER REVIEW

Neural restoration of greening defects in historical autochrome photographs based on purely synthetic data

S. N. Sinha, P. J. Kühn, J. Koppe, A. Kuijper, M. Weinmann

arXiv:2505.22291 · under review

[Paper](#)

Human-in-the-loop — curation of real data

GEM · 2026

Human-in-the-loop atlas-based 3D asset segmentation for interactive content workflows

P. J. Kühn*, **S. N. Sinha***, J. Hansen, R. Horst

*Accepted at GEM 2026 · *equal contribution*

[Paper](#)

PRE-PRINT · UNDER REVIEW

A human-in-the-loop deep learning framework for color reconstruction of lenticular films

S. N. Sinha, T. Kleist, G. Trumpy

arXiv:2608.02835 · under review

[Paper](#) [Project page](#)

Visual perception

WACV WORKSHOPS · 2026

RangeSAM: On the potential of visual foundation models for range-view represented LiDAR segmentation

P. J. Kühn, D. A. Nguyen, A. Kuijper, **S. N. Sinha**

Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) Workshops, pp. 1540–1548

[Paper](#)

ECCV WORKSHOPS · 2026

TinyDETR-Pose: Towards end-to-end real-time single-stage 6DoF object pose estimation with lightweight transformers

P. J. Kühn, D. A. Nguyen, **S. N. Sinha**, M. Weinmann, A. Kuijper

Accepted at ECCV Workshops 2026

[Paper](#)

ECCV WORKSHOPS · 2026

Optimizing multi-modal models for image-based shape retrieval: The role of pre-alignment and hard contrastive learning

P. J. Kühn, C. Spengler, **S. N. Sinha**, M. Weinmann, A. Kuijper

Accepted at ECCV Workshops 2026

[Paper](#)

Survey

JOURNAL · 2026

Computational game content generation and socially responsible design: Current practices, challenges, and open problems

R. Horst, R. Dörner, **S. N. Sinha**, P. J. Kühn

Computers & Graphics, p. 104683

[Paper](#)

References

Prof. Dr. Michael Weinmann e-mail: m.weinmann@tudelft.nl

Prof. Dr. Arjan Kuijper e-mail: arjan.kuijper@igd.fraunhofer.de